

Comprehensive Survey on Intrusion Detection Techniques for MANETs

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Abstract. Ad-hoc networks have been playing a vital role in various applications for many years. Dynamic Topology, Limited Resources, Open Environment, Lack of Infrastructure, Limited Bandwidth, Node Mobility are some of the key characteristics of the mobile adhoc networks and these unique characteristics post a significant threat to the security of the network. Encryption, authentication and firewall systems provide basic security features in Mobile Ad-hoc networks. However, the Intrusion Detection System (IDS) establishes extra security by preventing various attacks. Various IDS are built to provide better security features. In this work, the taxonomy of IDS based on different parameters are presented. This work compares performance of various IDS that are implemented using different techniques. This work concludes by indicating the research opportunities available for researchers to explore further.

Keywords: ad-hoc Network, intrusion detection system, manet, topology.

1 Introduction

The wireless networks entirely changed the way of communication in the world. Wireless networks use air medium to establish communication among nodes or devices. There are many wireless technologies available that include Bluetooth, Wi-Fi, 5G etc. [1, 2]. Applications of wireless technologies include surveillance and real-time multimedia applications, space and mission-critical applications and industry 4.0 applications. Wireless networks are divided into two parts, namely, Infrastructure-based Wireless Networks and Infrastructure-less Wireless Networks. The infrastructure-based wireless networks are dependent on a base station with a fixed number of nodes. They are expensive and they use predefined routes for data transmission. They cannot be used for hostile applications, such as earthquake prediction and military applications. Infrastructure-less networks do not have base stations and can be deployed without much infrastructure especially to mitigate natural calamities and disasters. Ad-hoc networks are classified into two classes based on mobility,

namely, Mobile Ad-hoc networks (MANET) and Wireless Sensor Networks (WSN). In MANETs, each node acts as both router and host, movable and does not rely on fixed position. But the nodes in WSNs are immovable and stay where they are installed. Various attacks extend from passive to active because of the open nature of wireless options. Attacks on these networks get affected from all sides, wired networks are guarded by gateways and firewalls. Threats may lead to damages which include node impersonation, data leaks, and message alterations. Some of the threat information is provided in Table 1

Table 1: List of Various Threats

Year	Threats	Description
Early 2000's	Hackers	Information is accessed by unauthorized users
2003	Intrusion Worms	Malicious software combining intrusion tactics and worm-like propagation, rapidly spreading across networks to exploit vulnerabilities and compromise systems
2005	Viruses, Spyware	Virus-Malicious software that infects the systems and self replicate Spyware-Gather sensitive information of users
2007	Botnets Spam	Steal data by spamming, DDoS attack or credential theft
2008	Malicious URLs	Phishing of messages which leads to security breaches
2011	Malicious Apps	Apps automatically connects to any services without our knowledge
2015	Ransomware	Malicious software encrypts a victim's files or locks them out of their system, demanding a ransom payment to restore access
2020 Onwards	Advanced Persistent Threats	Network attack done by hackers who get network gains.

In an ad-hoc network, a malicious node can cause several attacks which are very difficult to identify since tracking such nodes in a large number of nodes in an ad-hoc network is a difficult process.

2 Architecture of Intrusion Detection System (IDS)

Threats may infect systems or networks by breaking the primary line of protection i.e. privacy, routing, authentication, access control, key generation, and leaving the system or network unsafe [22]. MANETs are not able to identify any intrusions or attacks inside its network. The second line of protection includes an Intrusion

Detection System (IDS) which detects attacks with greater assurance and an Intrusion Prevention System (IPS) which stops them before causing big damage. IDS is a complete network that performs the following activities: *i) scans tasks in a single system or group of systems, ii) makes analysis of the inbound network traffic, iii) clashes incidents are filtered out which are against the system security policy, and iv) concludes the activity as an unauthorized activity from a malicious entity.* The objective of IDS is to automate the intrusion detection in the network that tries to interrupt the Confidentiality, Integrity, and Availability of the network i.e. CIA triad by any kind of attack.

Existing IDS cannot provide a good solution to all types of attacks and every system is constructed to provide a good security solution. Essential modules of IDS are data collection, data pre-processing, intrusion detection, and report generation as shown in Figure 1. The traffic data is collected from a single machine or network which consists of both normal and abnormal data. The collected data has unwanted and unordered data which should be cleaned and modified. The Intrusion Detection module is responsible for identifying malicious activities based on the rules provided in the knowledge base. IDS enhanced edition is provided by the administrator and contains a profile of already known intrusions and compares with the newly arriving data against the past intrusion profiles. In this way, any abnormal behavior from audit data can be identified. The report module generates the alarm when intrusion is detected. The nature of the attack is sent to the administrator and stored in the database for future reference. The performance measure of IDS is a difficult task due to many factors.

True Positive (TP): The identification is said to be true positive when intrusion is successfully identified. True Negative (TN): The identification is said to be true negative when it predicts the negative result correctly. False Positive (FP): In this state, the IDS detects it as malicious but the activity is legal. False Negative (FN): In this state, IDS shows the activity is legal but the activity is malicious. False Negative impacts MANET severely.

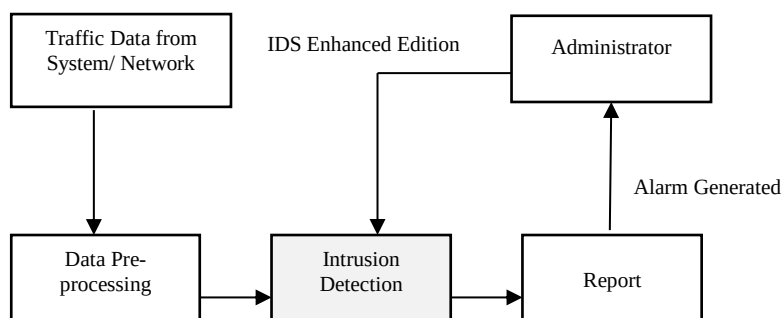


Fig.1. Intrusion Detection System

Table 2: Evolution and Timeline of Intrusion Detection System

Timeline of the IDS	Timeline	Architecture	Type of Attacks	Remarks
Neural Network & Watermarking Technique [6], 2007	2007	Self- Organizing Map	Routing behavior attacks	Additional training overhead leads to energy waste to find these attacks.
HIDS [22]	2008	Distributed and Collaborative	Black Hole Attack and Packet drop attacks	Test is needed before secure routing construction.
BHIDS- A black hole IDS [26]	2010	Hybrid	Black hole attack	Able to identify only black hole attack.
Leader Election Model [23]	2008	VCG model	Resource utilizing attack-selfish nodes	Selfish nodes do not drain energy to run IDS service, the percentage of packet analysis reduces with the help of time.
SCAN [24]	2002	Distributed and Collaborative	Routing misbehavior and packet forwarding misbehavior	Network services are temporarily halted during detection.
A System of Umpires [25]	2010	Collaborative IDS for double and triple users	Packet dropping attacks	Umpires are not permanent. Election is needed to re-elect the leader. Energy consumption is high.
Feed-forward neural network, Support vector machine. [27]	2016	Distributed and Collaborative	Grey Hole and Rushing Attacks	Fuzzy Petri Nets can be added to enhance the performance
Adaptive Neuro Fuzzy Inference System, Particle Swarm Optimization [28]	2020	Hybrid	Black hole attack	Able to identify only black hole attack.
Proportional Overlapping Scores (POS)	2022	Collaborative	Rushed Black and Gray Hole attacks	Node mobility is not considered.

History of IDS started three decades ago with the trend of network usage by users. The term Intrusion Detection was introduced by James P. Anderson in the U.S Air force during the 1980s. He had introduced automated IDS reports with help of Computer Security hazard Surveillance and Monitoring. The first application built method is used as an anti-virus. Some important issues encountered by IDS in ad-hoc networks are discussed and these challenges must be resolved. The frequent changes in topology offer IDS with flexible architectures. It is desirable to develop co-operative and distributive IDS due to lack of central management. Ad-hoc networks are vulnerable since they lack proper boundaries and allow outside attacks. The bandwidth of Ad-hoc networks is limited. This forces the IDS to minimize the interaction between the nodes as much as possible. IDS has to follow cryptography since the information should not be revealed easily when agents are compromised. The power source is limited in ad-hoc networks. So, IDS uses a low computational approach. However, the initial IDS has few problems. i) It could send alert messages only for known attacks; ii) resources are drained frequently due to continuous scanning and updates of the signature list. Table 2 describes the evolution of IDS.

3 Classification of Intrusion Detection Techniques

Many methodologies are used to detect intrusions in mobile ad-hoc networks. This section deals with the most important and well-known techniques that are used to find the intrusions. These techniques include signature-based, anomaly based, and hybrid based. Figure 2 shows the classification of intrusion detection systems.

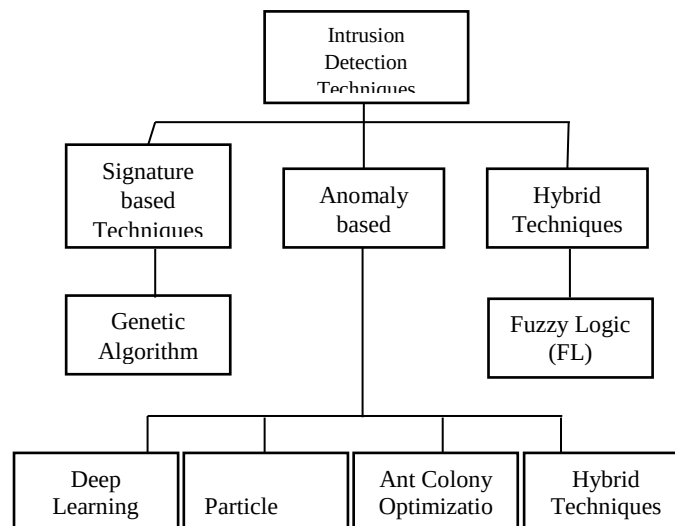


Fig. 2 Classification of Intrusion Detection System

3.1 Signature based IDS

A signature is a pattern of a unique network traffic change. All the patterns are stored in the database and the new patterns are reviewed with the previous records. Signature-based techniques are called as knowledge-based technique as it uses previously recorded patterns to check the intrusions. It works like an antivirus which is based on the recorded virus patterns but fails in detecting new threats.

3.1.1 Genetic Algorithm based Intrusion Detection

Natural selection is the base for genetic algorithms. It simulates the survival of the fittest or natural selection process. A genetic algorithm selects individuals from random population and produces the children for the next generation. Shingo et. al have proposed a network intrusion detection model based on fuzzy class association rule mining using Genetic Network Programming (GNP) [3]. This method finds the differences between the behavior of the unauthorized connection and the normal connection using a fitness function. This technique is progressively optimized and uses direct graph structures. It contains databases that include both discrete and continuous attributes. Association rule is generated based on the nature of the attack to identify threats. The methodology proposed in [3] captures both misuse and anomaly types of attacks. Genetic algorithm and Support Vector Machine (SVM) are the building blocks of the proposed IDS. It uses feature selection, weight, and parameter optimization for effective classification by SVM. Crossover and mutation probability of GA is applied for optimization. Fitness value and population evolution algebra are used for effective feature selection that reduces error rate, increases true positives, and reduces classification time subsequently.

3.2 Anomaly based IDS

This technique is also called behavior-based IDS. The normal behavior is recorded and derived over time by monitoring various activities such as hosts, network connection and regular activities. The specific threshold is marked and any deviation from this value is called an anomaly. An alarm is generated when there is a large amount of deviation. It is preferred over signature-based models since the new intrusions are detected effectively. A profile is created from attributes such as the usage of the processor, the number of emails sent and the number of login attempts failed. The model has two phases: training and detection. In training, the behavior is learnt. When any intrusion occurs, the IDS checks with the usual behavior. If a particular behavior does not match with a normal profile, it alarms and marks the behavior as abnormal. The unknown behaviors are alarming and they are sent to the expert team and stored in the database for future reference.

5.2.1 Deep Learning based Intrusion Detection

Sydney et. al have proposed an intrusion detection system that uses feature extraction with deep learning [7]. There are two types of feature extraction, namely, wrapper-based and filter-based. Deep Learning (DL) is used in wrapper-based feature extraction. It uses DL models to classify and measure the performance of those features. Feed-forward deep neural networks are used to produce a reduced optimal feature vector. The proposed model in [7] is applied for both binary and multiclass classification. Mohammad et. al have

proposed an intrusion detection DL model for a big data environment using a hybrid deep learning approach [8]. Models such as Convolutional Neural Networks (CNN), Deep Brief Networks (DBN), and Weight-Dropped Long Short-Term Memory (WD LSTM) networks are used to detect intrusions. CNN extracts and gathers information about the meaningful features from the data traffic present in a network that exploits the speed. LSTM helps to retain the dependencies that are long-term and prevents the gradient vanishing problems. DL models are evaluated with the help of intrusion datasets such as UNSW-NB15 and ISCX2012 [10]. Wisam et. al [9] have proposed network intrusion detection for deep learning architecture using a double PSO metaheuristic. More deep learning models are pre-trained by a double PSO-based metaheuristic algorithm to achieve high performance. This revised algorithm consists of two different levels upper and lower levels. The upper level is used to select optimal features from a given subset and later hyper-parameter is used to increase detection accuracy from a reduced data set. The proposed model in [9] used the Friedman statistical test and two different ranking methods to conduct the test. The detection accuracy rate increased from 4% to 5% and the false alarm decreased. Arwa et. al have proposed a detailed survey on anomaly-based IDS using deep learning approaches [10]. The proposed methodology in [10] provides a taxonomy of deep learning for IDS with different aspects. Zhendong et. al [11] have proposed an IDS model using Semantic re-encoding [11]. It increases the semantics between the traffic so that the abnormal traffic can be easily distinguished and identified. The robustness and performance of the models are improved by using the proposed methodology in [11] and it identifies web injection attacks effectively. A survey and taxonomy of deep learning models are presented by Sunanda et. al in [12]. Models such as feed-forwarding neural networks, LSTM networks, autoencoder, and DBM are used for evaluation. Four datasets i.e. Two legacy KDD 99, and NSL-KDD and two modern CIC-IDS2017 and CIC-IDS2018 are used for training. Mohamed et. al have proposed an IDS by implementing DL models that are trained by 35 well-known cyber datasets which are categorized into seven categories, namely, Network traffic-based, Internet traffic-based, Virtual Private Network-based, Electrical network-based, Android apps-based, IOT traffic-based and Internet-connected devices-based datasets [13]. These seven deep learning approaches come under two models such as deep discriminative models and generative unsupervised models. The robustness and accuracy of IDS is improved by using these different DL models.

3.2.2 Particle Swarm Optimization based Intrusion Detections

Particle Swarm Optimization (PSO) is a method that effectively optimizes a problem just by trying to improve the solution of a candidate with respect to a given measure in an iterative way. Botnet threat is a huge risk for internet users. The BD-PSO-V model is proposed in [14] to detect botnets using a hybrid technique. This technique is a combination of particle swarm optimization and voting system. PSO is used to select appropriate features and a voting system is used for the classification of samples. Datasets such as ISOT [4] and Bot-Iot [5] are given as inputs for PSO and Deep Neural networks, and SVMlib, and Decision tree are

given as inputs for the voting system. Decision-making is done based on maximum votes. Hadeel Alazzam et al. [15] have proposed wrapper method-based feature selection to identify intrusion detection using an inspired Optimizer (PIO). A robust IDS is developed with a reduced feature set. Discrete optimization process i.e. cosine similarity is applied to swarm intelligent algorithms to solve problems. Bach Hoai Nguyen et al. [17] have proposed a new survey on feature selection in classification problems using swarm intelligence. Swarm intelligence adopts an easy mechanism to filter relevant data from a large data set. This survey considers three SI algorithms, namely, ABC, PSO, and ACO.

3.2.3 Ant Colony Optimization based Intrusion Detections

The Ant Colony Optimization algorithm is a technique used to solve computational problems. This algorithm is used to reduce the computations by using graphs. Artificial ants are made as agents mimicking the behavior of original ants. Ahamed Fayez Tuani et al. [18] have proposed solutions for the traveling salesman problem using heterogeneous adaptive ant colony optimization (ACO) with a 3-opt local search. Good parameters are chosen by adaptive tuning methods that include ant colony optimization to reach near-optimal solutions. Fitness can be improved by a 3-opt local search heuristic strategy. Yassir Aberni et al. [19] have proposed a palm vein recognition method for personal authentication and identification. This is based on the scheme of competitive coding which uses a Multiscale Local Binary Pattern (MLBP) along with Ant Colony Optimization (ACO).

3.2.4 Reinforcement Learning based Intrusion Detection

Reinforcement Learning based Intrusion detection system helps in identifying the network layer attacks happening in uncertain environments. Double layer attack detection framework based on Deep Q- learning and Deep Autoencoder to detect False data injection attack. The algorithm produces 98.20% detection rate and False Positive rate of 2.9%. Reinforcement based malicious node [30] is devised to mitigate network layer attacks in adhoc networks. The proposed method outperforms the state of art techniques in terms of Accuracy and False positive Rate. Reputation based opportunistic routing method using Q-learning [31] is developed to mitigate the network layer attacks such as black hole and grey hole attack. This method is efficient in terms of packet loss, average end to end delay and energy level. Q-learning based technique is proposed for secured routing [] in On demand Distance Vector protocol which enables the NS-3 simulator to consider more number of neighboring nodes for simulation. This technique makes the network more reliable by detecting the black hole attack with high accuracy.

3.3 Hybrid Intrusion Detection System

Fuzzy logic allows processing of multiple truth values using a single variable. It also exists in the real biosphere. It is implemented in Intrusion Detection Systems as it has the ability to merge the data of different sources frequently. Selvakumar K et al. [29] have proposed IDS which is based on the Fuzzy rough set-based nearest neighbor classification. This

model is used to train the classifier. The number of features is to balance between optimal detection rate and performance. Allen's intervals of algebra operators removed the redundant features. False positives and false alarm rates are reduced in this model. Intrusion detection is developed for real-time using neuro-fuzzy classifiers and implemented on GPUs [30]. A high detection rate is achieved by an Enhanced fuzzy min-max neural network. Minimum training and recall time are enough. Manimurugan et al. [31] have proposed an IDS which is based on the crow search optimization algorithm which is in combination with an Adaptive neuro-fuzzy inference system. Here the fuzzy rules are not imposed in the input information and the hybrid learning uses the data pairs of input and output. The list of datasets that are used for IDS is listed in Table 3.

Table 3: Overview of IDS Classification Mechanisms

Primary IDS Techniques Used	Additional Schemes used	Parameters Considered
Genetic Algorithm [3]	GNP-based Fuzzy Class Association Rule	Detection Accuracy, Low False Alarm Rate
	HNGFA Framework, OGFA & IGFA	NIDS Detection Accuracy
Deep Learning [7]	IFS, CNN Extractor & BG Classifier and 5-fold CV	BG Classifier Over fitting Problem & Performance
	Feed Forward Deep Neural Networks, Extra Trees (ET) algorithm, WFEU-FFDNN, CNN-WDLSTM IDS Large-scale big data	Binary and Multi-class classification, NIDS and real-time IDS
	SRDLM method for re-encodes, SRDLC	Accuracy and Robustness of algorithm, Web character injection network attack
Particle Swarm Optimization [14]	WOA and genetic operators Performance	Binary and Multi-class classification
	Friedman statistical test & Ranking methods	BD-PSO-V Botnet Threat
Ant Colony Optimization [17]	3-opt local search	Multi-scale local binary pattern and Low and small budget function
Fuzzy Logic [19]	Allen's intervals algebra operators	Redundant data removal
Reinforcement Learning	Deep Q-Learning with Deep Autoencoder	Detection Rate, False Positive Rate
	Reinforcement learning-based malicious nodes	Accuracy and False Detection Rate
	Reputation Opportunistic Routing based on Q-learning (RORQ)	Packet loss, Average end-to-end delay, Energy efficiency
	Q-learning in Ad-hoc On-demand Distance Vector Protocol (AODV)	Accuracy and Packet Drop Ratio

Table 7: Important Dataset for IDS

Dataset ID	Description
UNSW-NB15	Network traffic dataset for intrusion detection, focusing on network-based attacks.
AWID	Ambient Wi-Fi and Bluetooth device dataset for activity recognition and localization research.
CICIDS2017	Intrusion detection evaluation dataset from the Canadian Institute for Cybersecurity, 2017 edition.
ADFA-LD	Australian Defence Force Academy Labeled Dataset, used for intrusion detection and network security research.
KDD	Knowledge Discovery in Databases (KDD) Cup 1999 dataset, widely used for network intrusion detection research.
NSL-KDD	Network Security Lab Knowledge Discovery in Databases (NSL-KDD)

4 Conclusions

The paper presents a survey on the features of Mobile Ad-hoc Networks and their challenges. Various types of attacks, such as active attacks, passive attacks, wormhole, and black hole attacks, occur in different scenarios within ad-hoc networks. The Intrusion Detection System (IDS) is elaborated, including its working modules, challenges faced, requirements, and its role in ad-hoc network security. Detection techniques, datasets used, associated schemes, tools, merits, and demerits which are supported by figures and tables are present in this survey. The interactive architecture and key information of IDS types have been explained based on the detection mechanisms provided by the taxonomy of IDS. Classification of IDS includes Signature-based, anomaly-based, and Hybrid techniques. IDS detection methodologies containing various schemes are summarized. Many issues in previous works related to IDS are pointed out and elaborated. Strong points and limitations are clearly stated for the work of each reference. Clear comparisons are given through detailed tables.

In the future, it would be a great idea to increase the range of different attacks in a single attempt and consider privacy issues when installing the system in a mobile ad-hoc network. Dealing with a multi-level or hybrid approach poses a high computational cost for attaining results. Future research aims to adopt mechanisms that achieve cost-effectiveness. Shortening the expected time for feature training and testing is crucial. Reducing false alarms plays a significant role in IDS since these issues degrade overall performance. Existing intrusion detection systems lack sufficient accuracy in real-time threat detection. Thus, new research focuses on real-time IDS to identify attacks promptly.

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